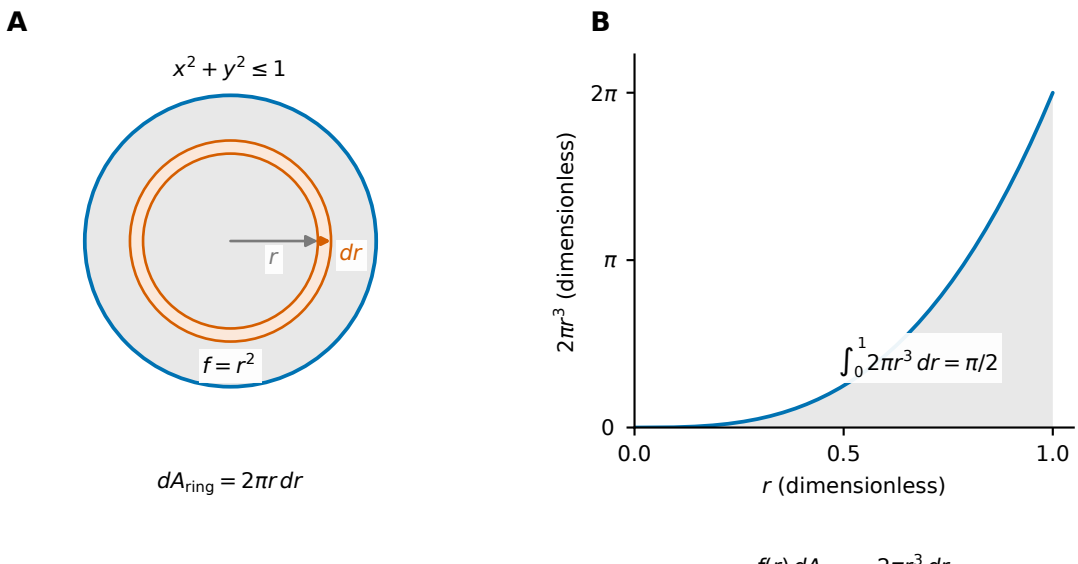
A
$x^2 + y^2 \le 1$
r
dr
$f = r^2$
$dA_{\text{ring}} = 2\pi r\, dr$
B
2π
π
0
$2\pi r^3$ (dimensionless)
$\int_0^1 2\pi r^3\, dr = \pi/2$
0.0
0.5
1.0
r (dimensionless)
$f(r)\, dA_{\text{ring}} = 2\pi r^3\, dr$